SFB 1252 PROMINENCE IN LANGUAGE

On prosodic focus marking in German declarative questions

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Focus structure, context, and prosody

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- The focus structure of assertions can be controlled via context questions:

(1) Q: Who did Mary greet? A: Mary greeted [**Peter**]$_F$.

(2) Q: Who greeted Peter? A: [**Mary**]$_F$ greeted Peter.

(3) Q: What did Mary do to Peter? A: Mary [**greeted**]$_F$ Peter.

(4) Q: What happened? A: [Mary greeted **Peter**]$_F$.

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Declarative questions

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(5) Maria hat Peter gegrüßt. (declarative assertion)
(6) Maria hat Peter gegrüßt? (declarative question)
(7) Hat Maria Peter gegrüßt? (V1 interrogative)

- Declarative questions:
  - String-identical to declarative assertions
  - Prosody and/or context disambiguate the intended speech act
  - Generally have rising contours
  - Nuclear accent often has a low or bitonal pitch target

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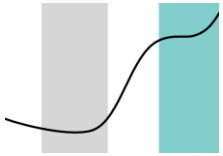
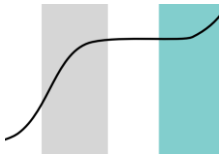
Nuclear contours of (some) German questions

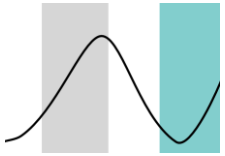
- GToBI describes 2 main rising contours for questions:



L* H-^H%



L+H* H-^H%



L+H* L-H%

Graphs are from www.gtobi.uni-koeln.de

Nuclear contours of (some) German questions

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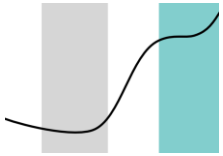

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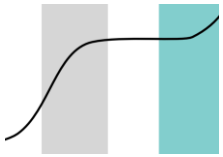

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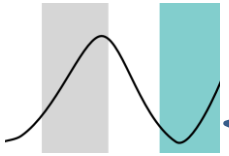
Simple rise



L+H* L-H%

Fall-rise

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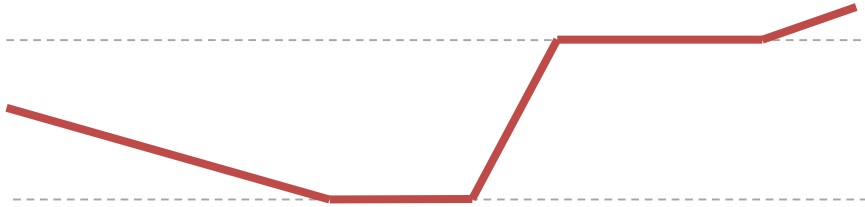
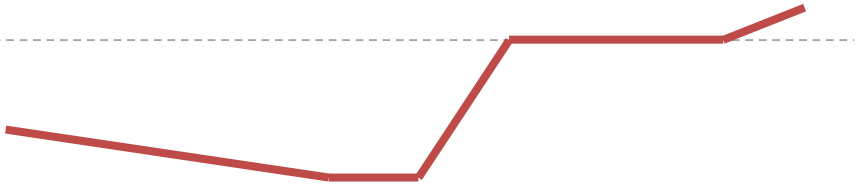
Research questions



Maria hat [**Pe: ter**]$_F$ gegrüßt?



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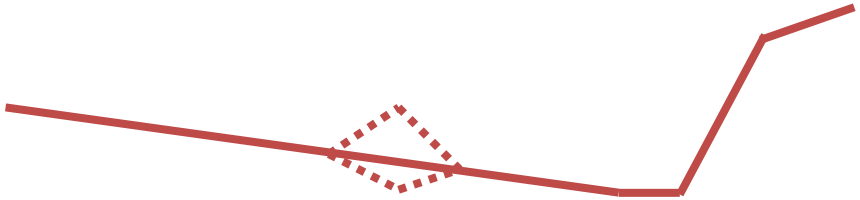

Maria hat Peter [**gegrüßt**]$_F$?

Top:
1. Are there phonetic differences between narrow and broad focus on the object?

Left:
2. Does narrow focus on the verb shift the location of the nuclear accent?
3. Are there pre-nuclear accents, and what forms do they take?

Production study: design

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- Design: Pseudo-dialogs between 3 speakers
- 2nd speaker introduces a supposition, which is then questioned by 3rd speaker (= participant)
- 4 conditions:
  1. Object new, verb new
  2. Object given, verb given
  3. Object narrowly focused, verb given
  4. Object given, verb narrowly focused
- 24 speakers, 12 items; 1138 utterances analyzed

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Example item: new and given conditions

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S1: Übrigens, Jana hat mir am Montag erzählt, dass sie diesen Sommer ein Praktikum im Schwarzwald machen will. Da möchte sie wohl bei 'nem Jäger mehr über die ganzen Waldtiere lernen.

Object and verb both NEW

Object and verb both GIVEN

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S3: **Jana will da dann Rehe jagen?** Interessant. Was man im Leben nicht so alles macht.

Object and verb both NEW

Object and verb both GIVEN

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Example item: narrow focus conditions

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S3: **Jana will da dann Rehe jagen?** Hasen findet sie wahrscheinlich zu niedlich. Das könnte ich verstehen.

S3: **Jana will da dann Rehe jagen?** Erforschen ist ihr wahrscheinlich nicht aufregend genug. Jana lässt echt nichts aus.

OBJECT focused

VERB focused

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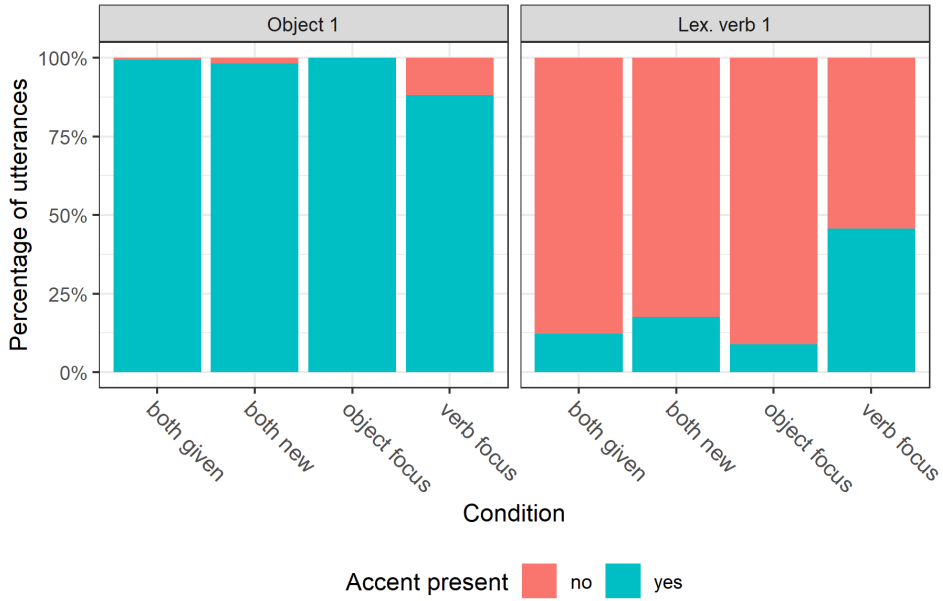
Overall results



Condition	Object accented	Verb accented
Both given	99%	12%
Both new	98%	18%
Object focus	100%	9%
Verb focus	88%	**46%**

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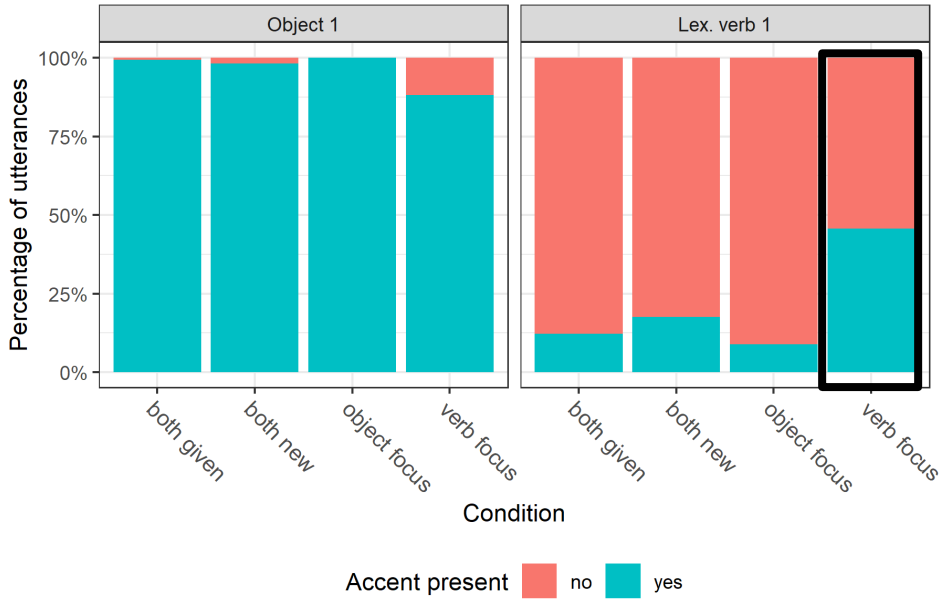

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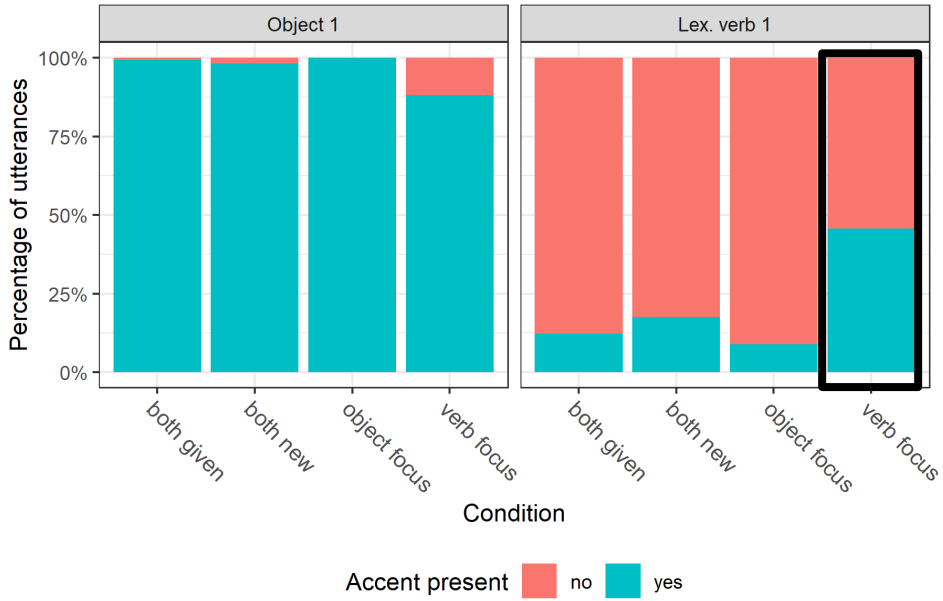

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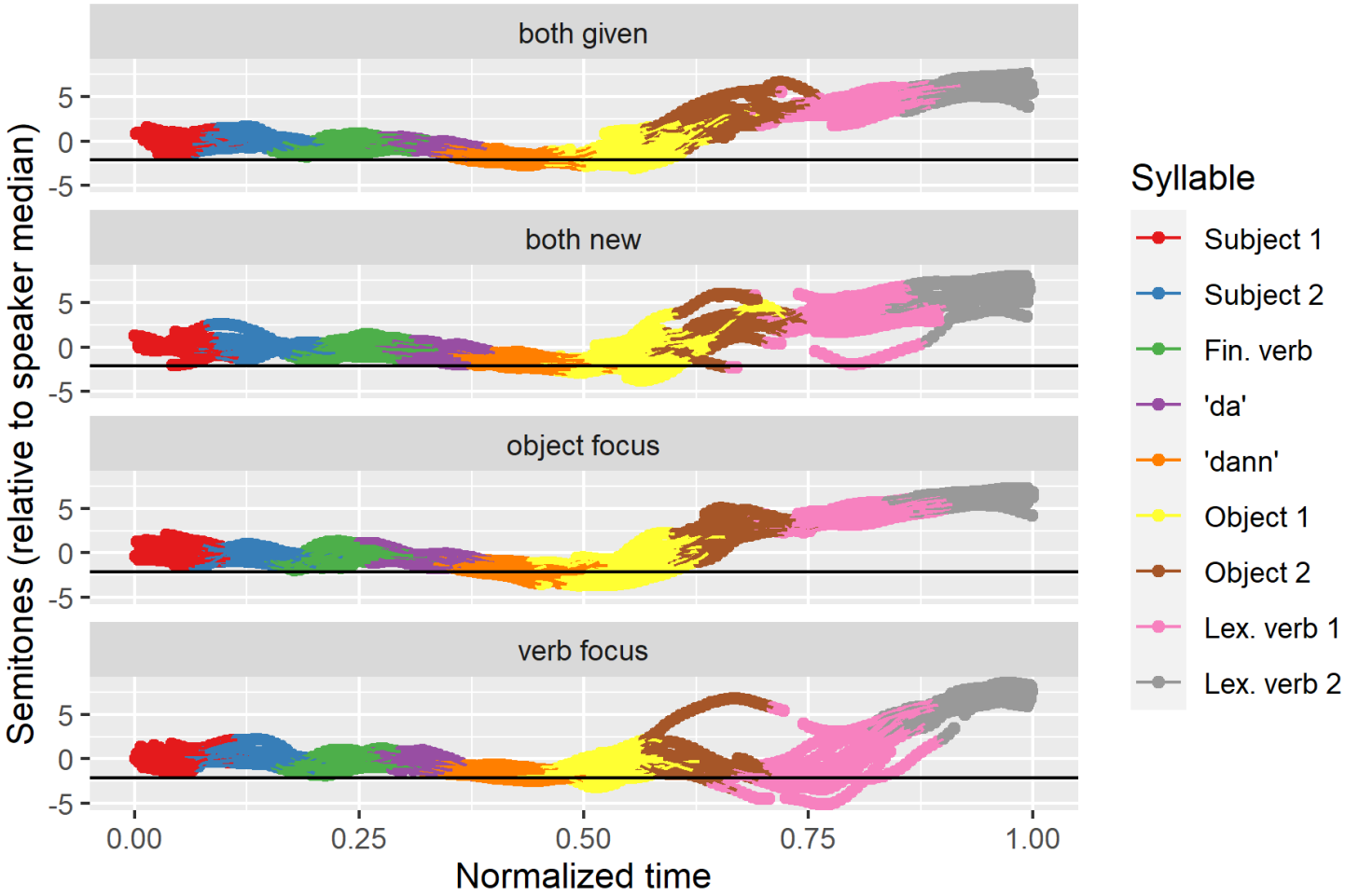
- Narrow focus on the verb: increase in number of verb accents, slight decrease in object accents
- 46% of utterances with focused verbs have a verb accent
- But this number hides substantial **inter-individual variation**

A participant who realized differences between the conditions



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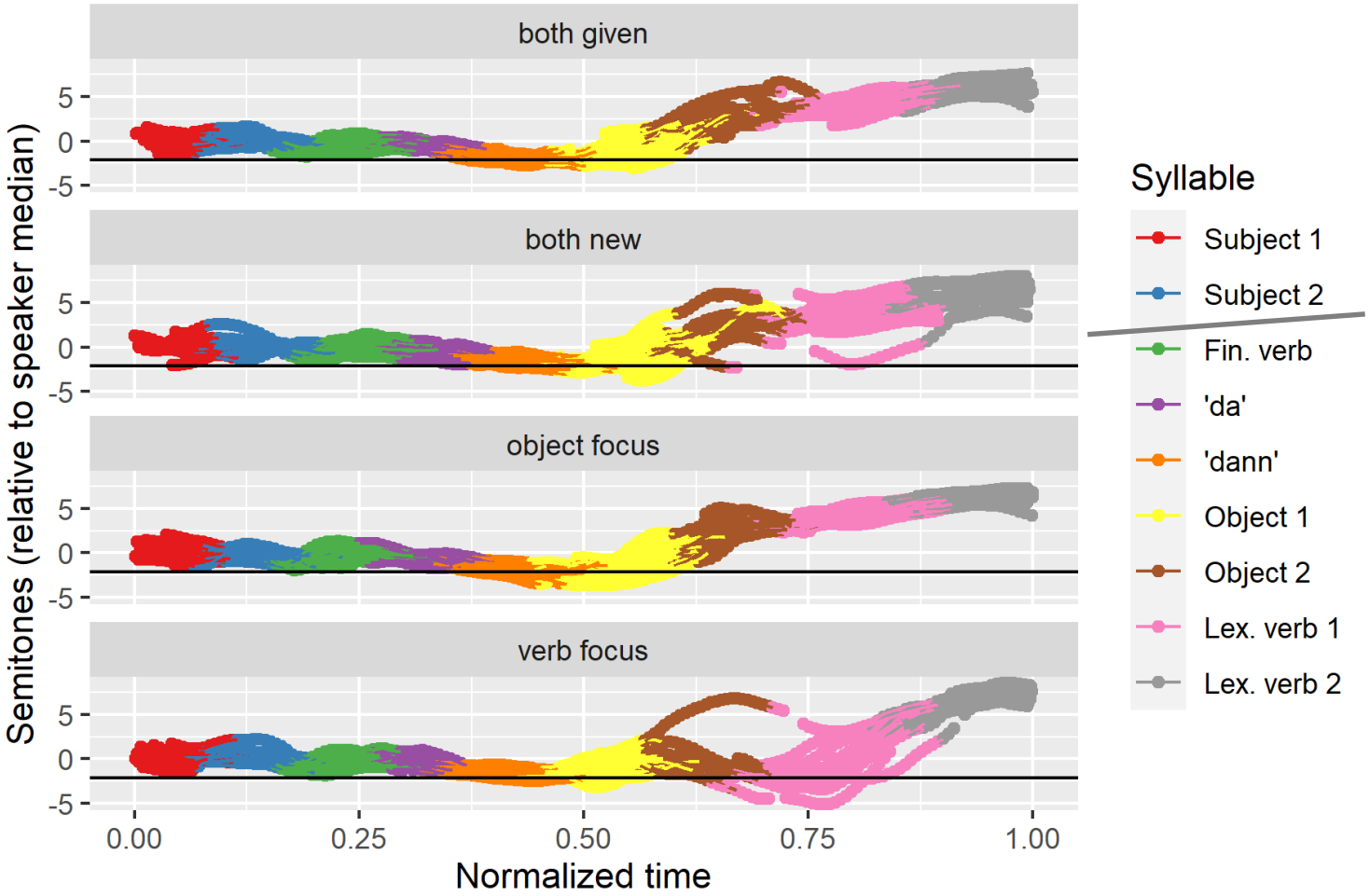
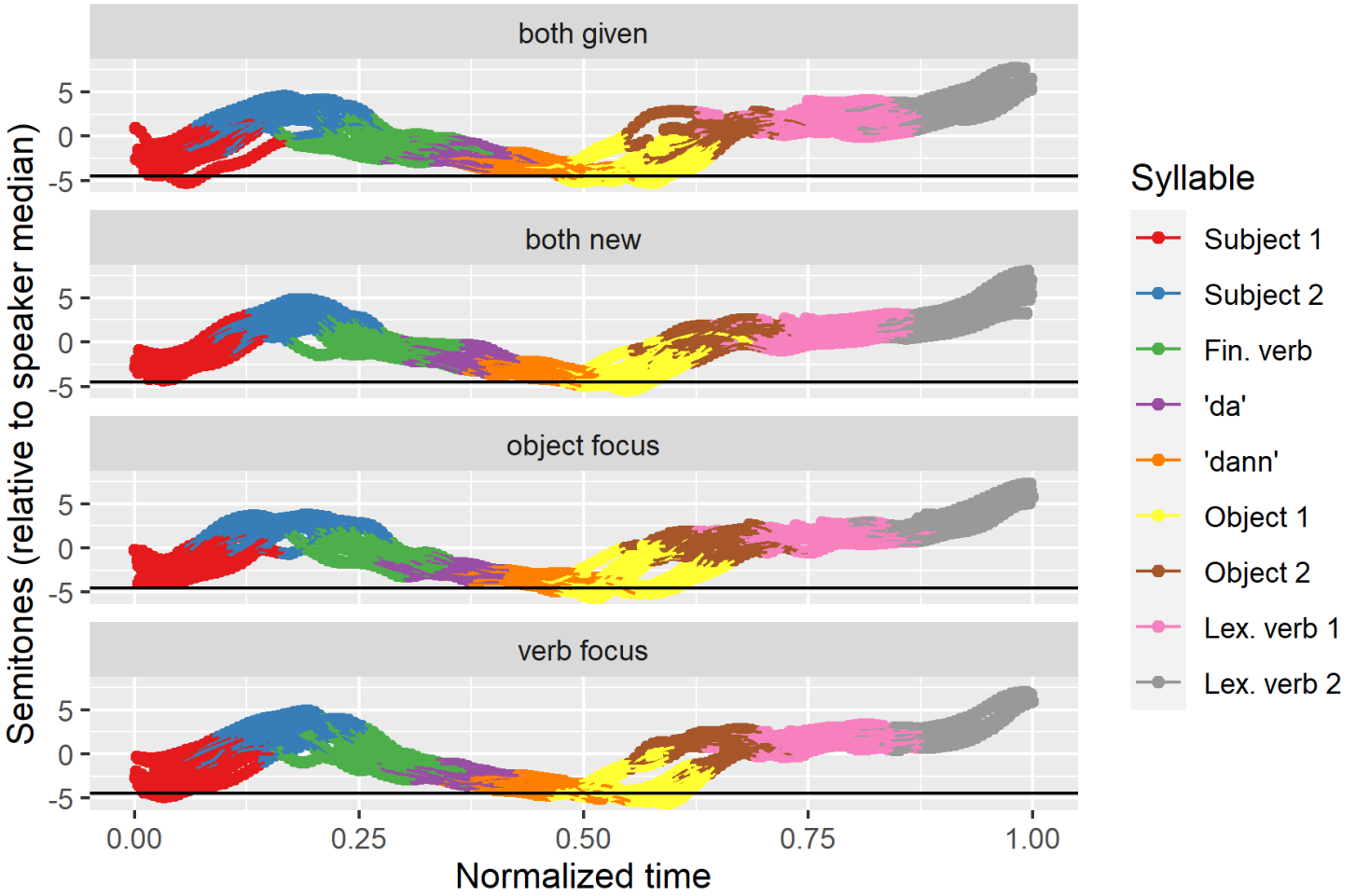
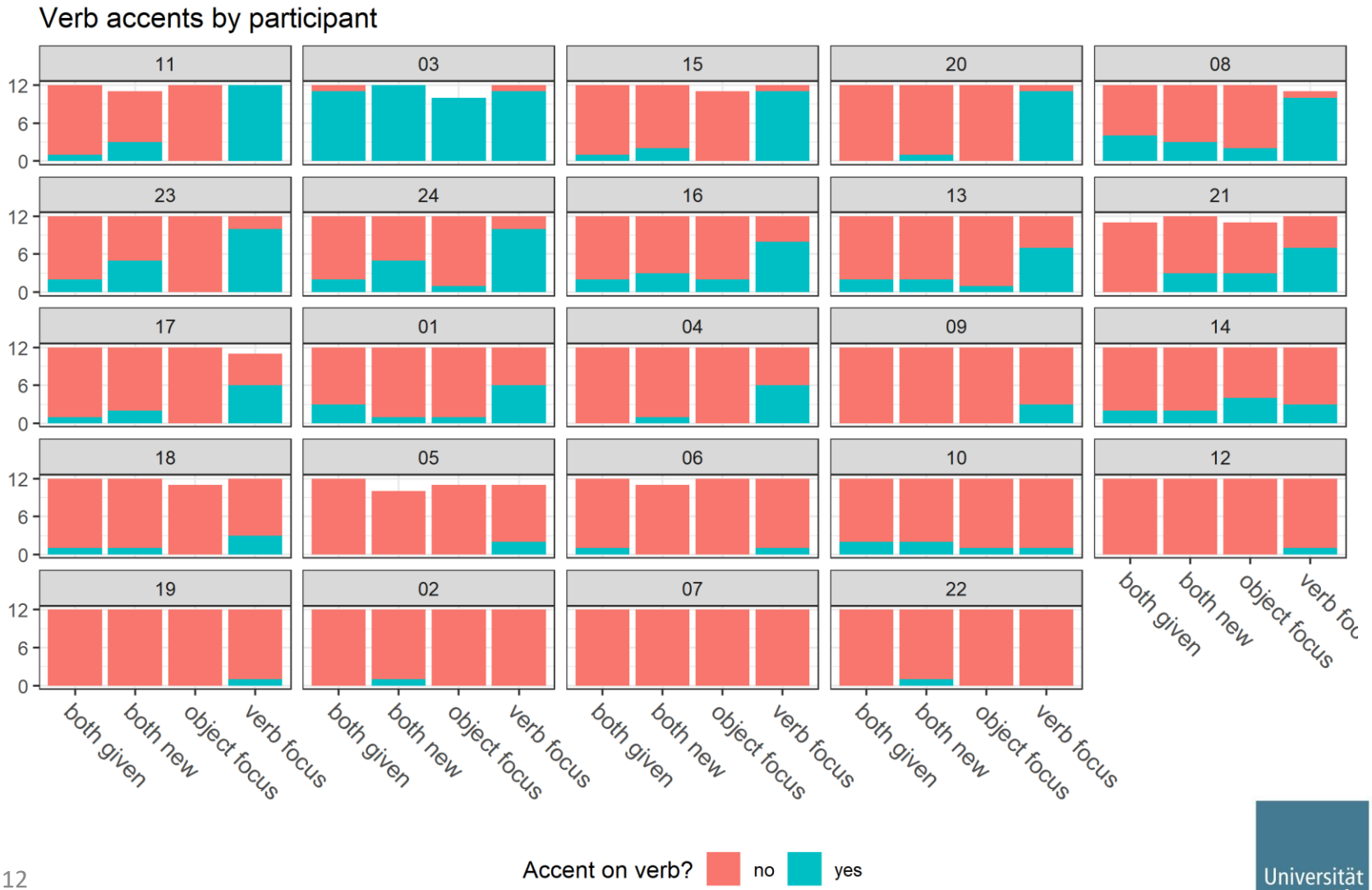


Black line: average minimum pitch on the object in the "both new" condition

A participant who didn't show differences



Inter-individual variation in verb accents



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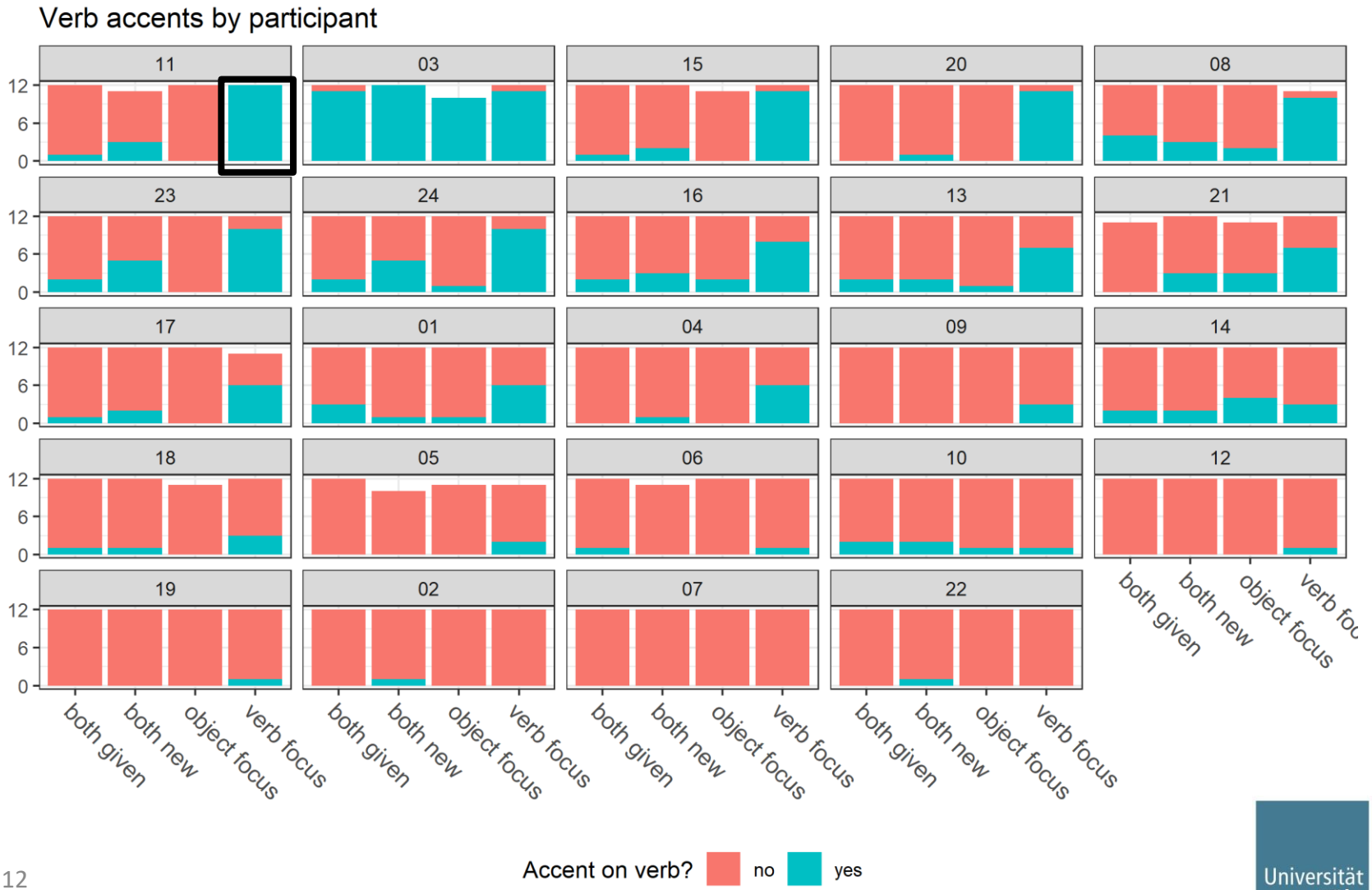

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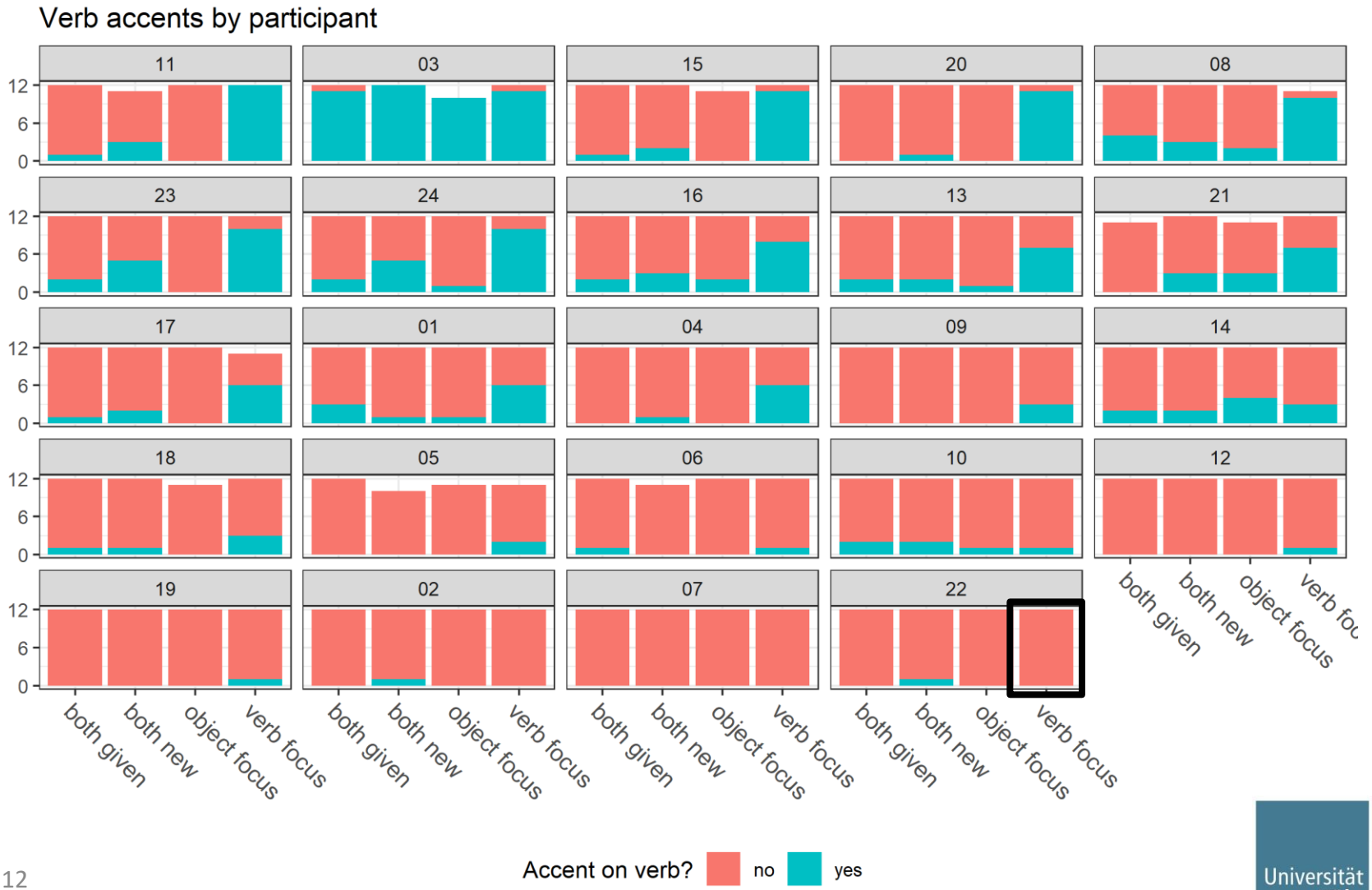

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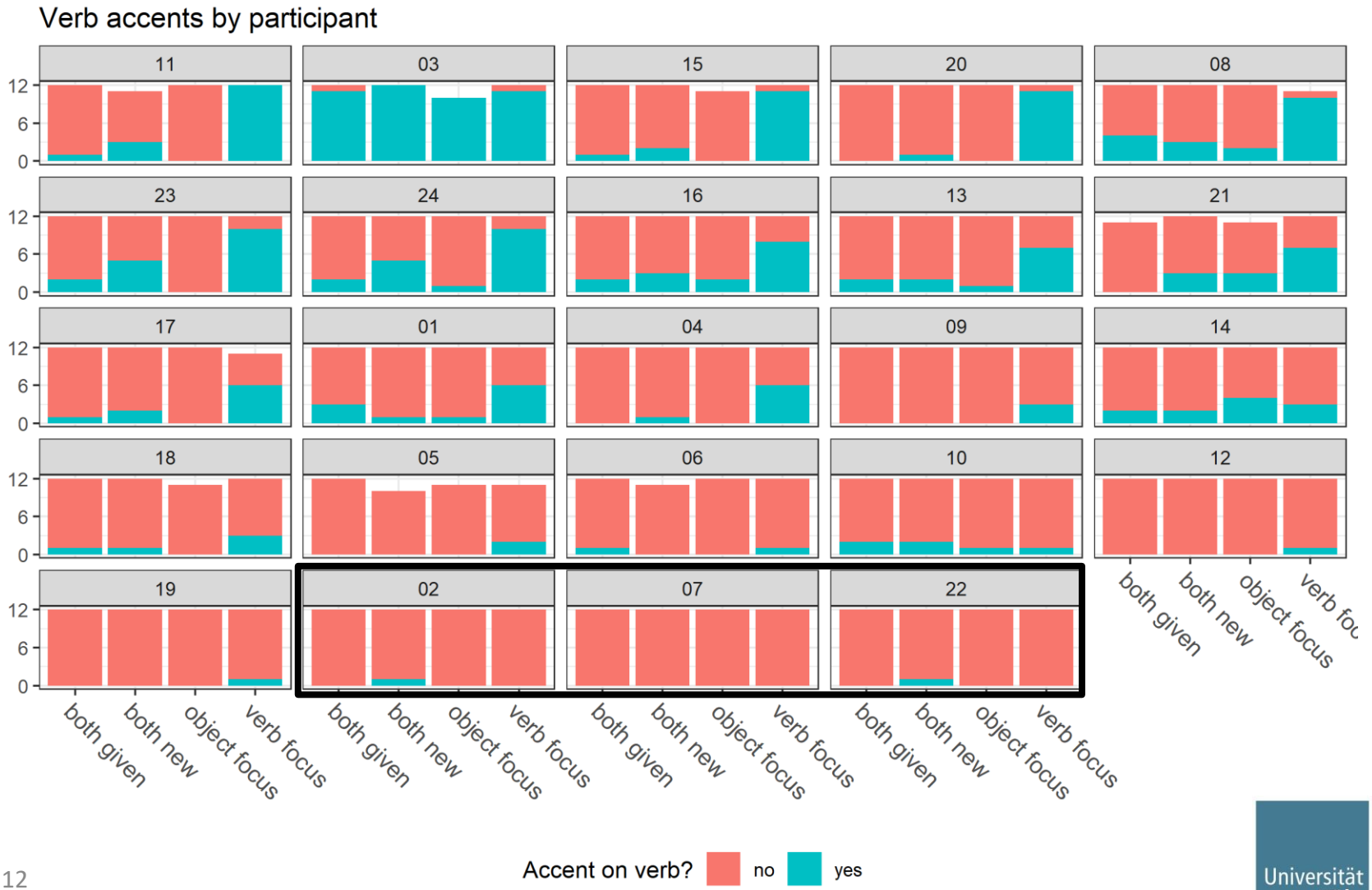

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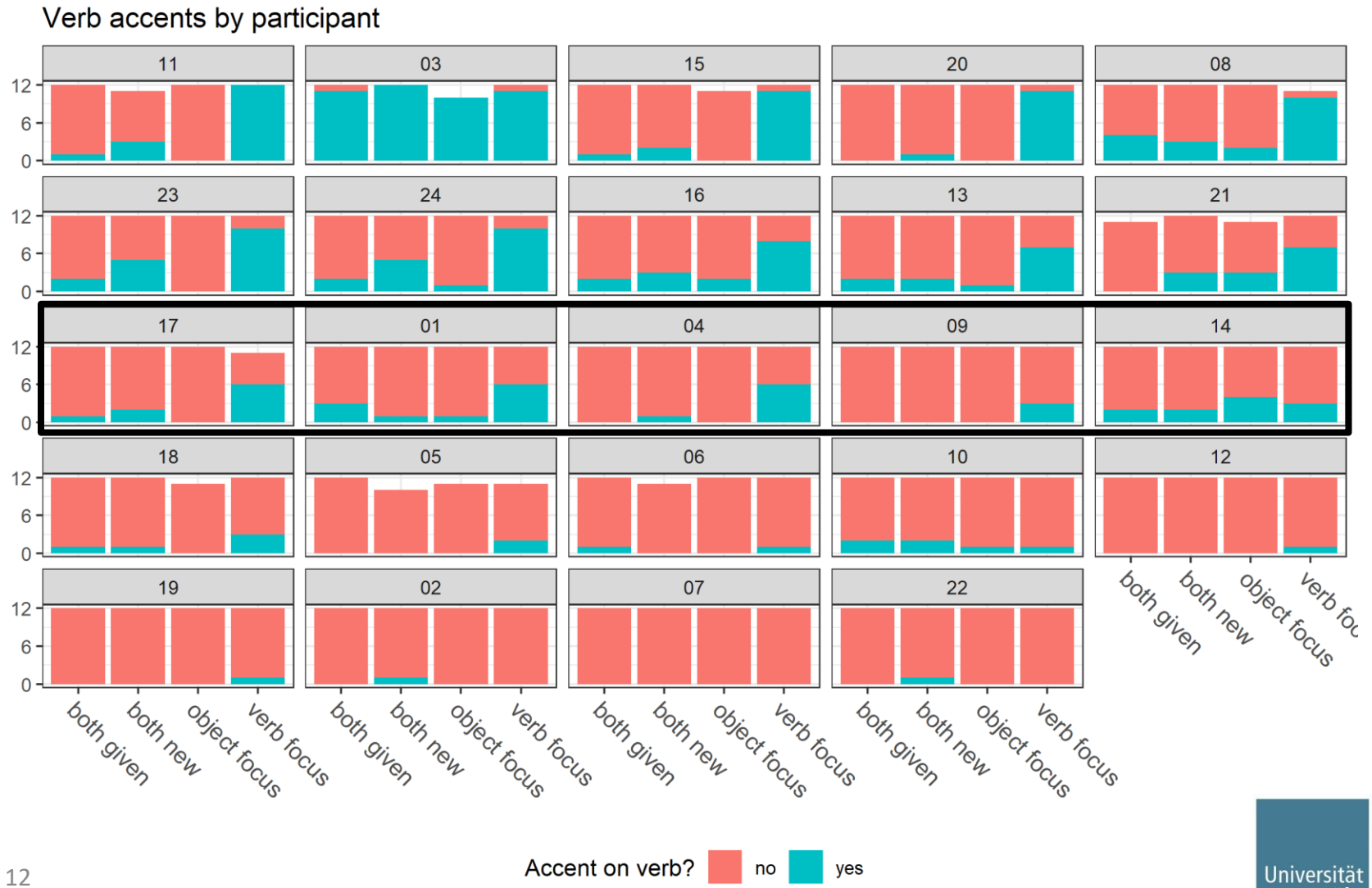

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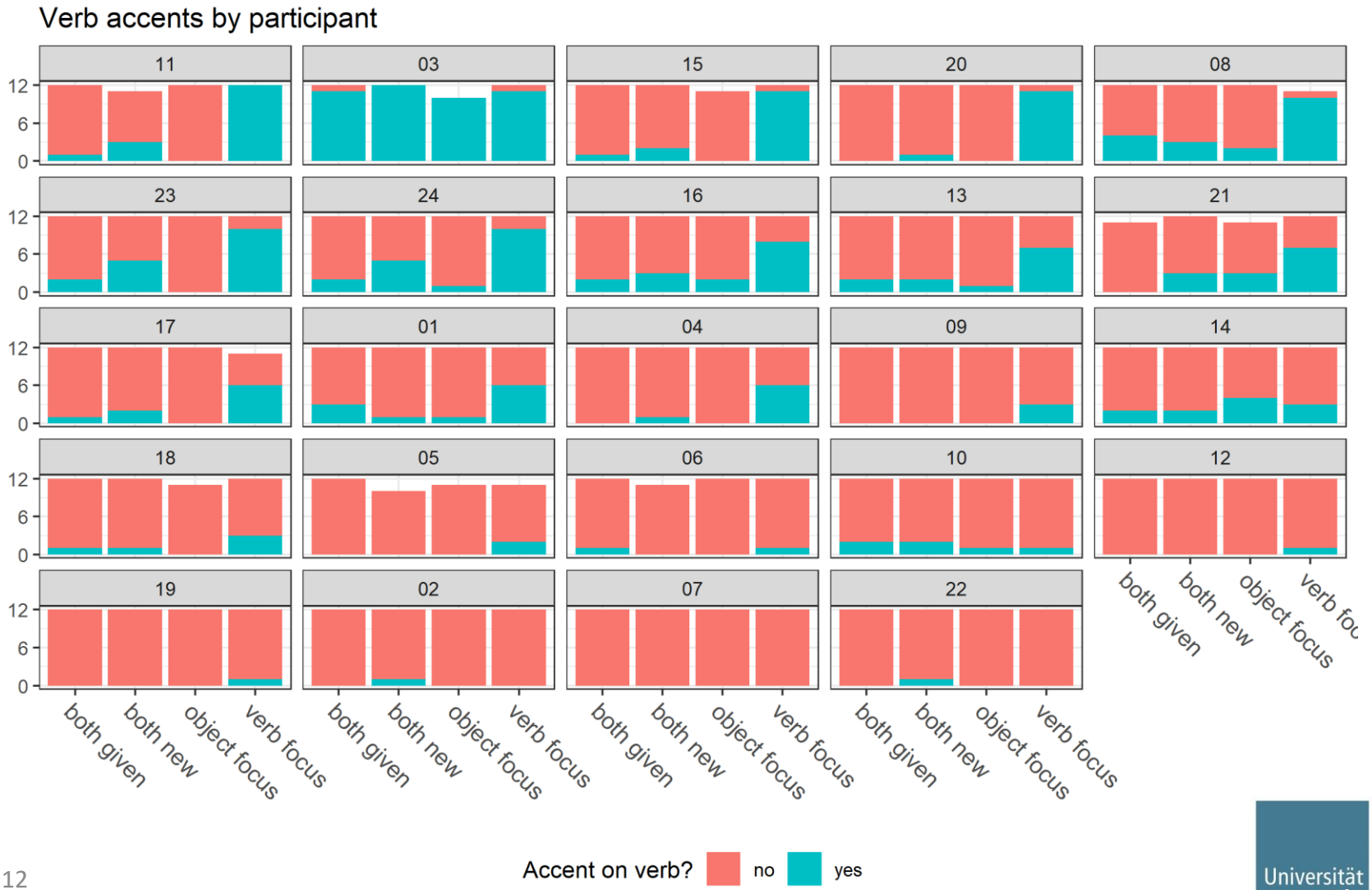
Inter-individual variation in verb accents



Post-hoc check: participant sub-groups

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- Participant sub-groups with respect to **accent placement on focused verbs**:
  - Often or always: more than 66.6% of utterances have an accented verb (7 participants)
  - Sometimes: 50% or more (5 participants)
  - Rare: less than 50% (8 participants)
  - Never: 0% (3 participants)
  - (1 special case: verb almost always accented regardless of condition)

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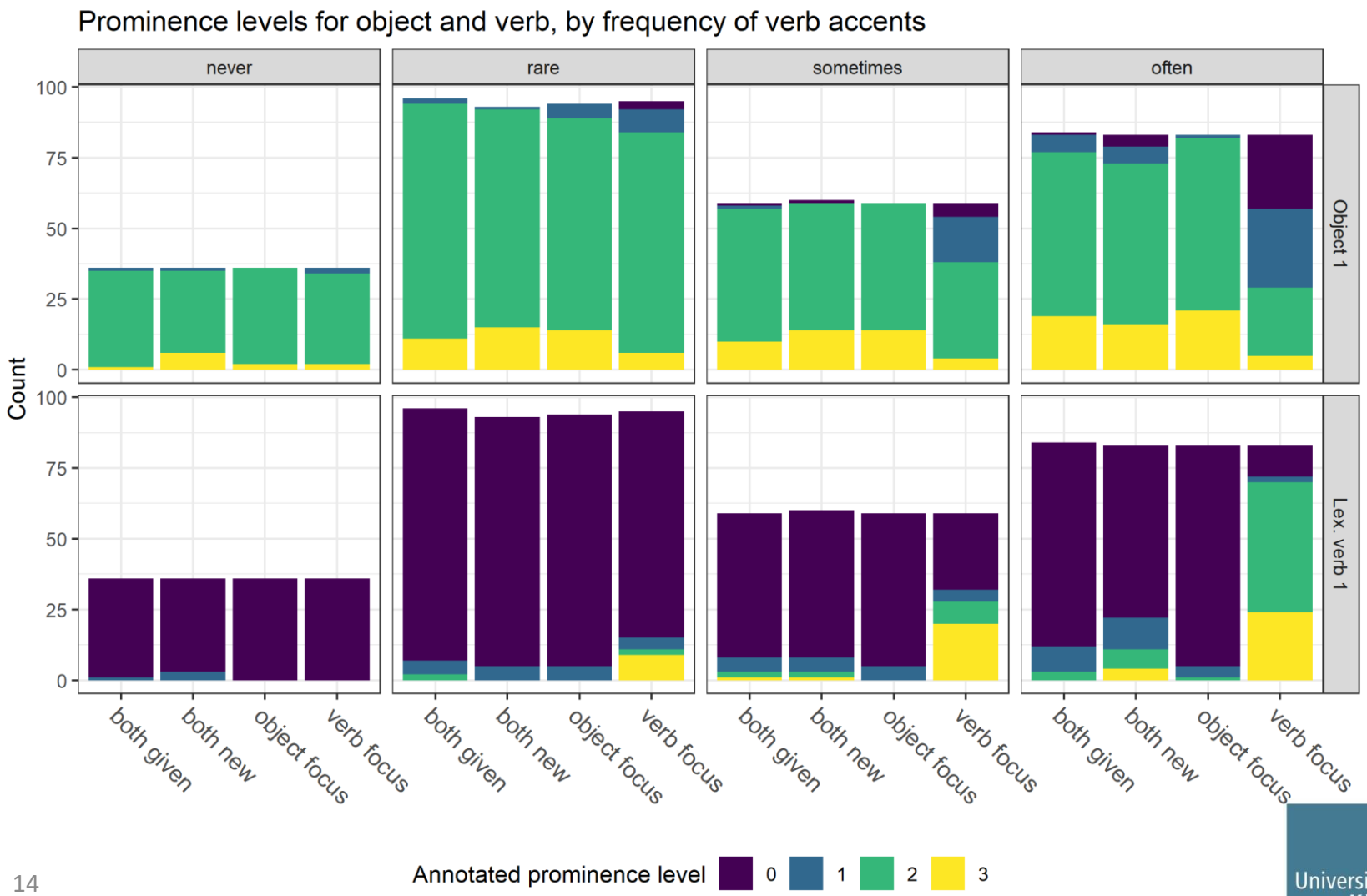
Prominence by sub-group



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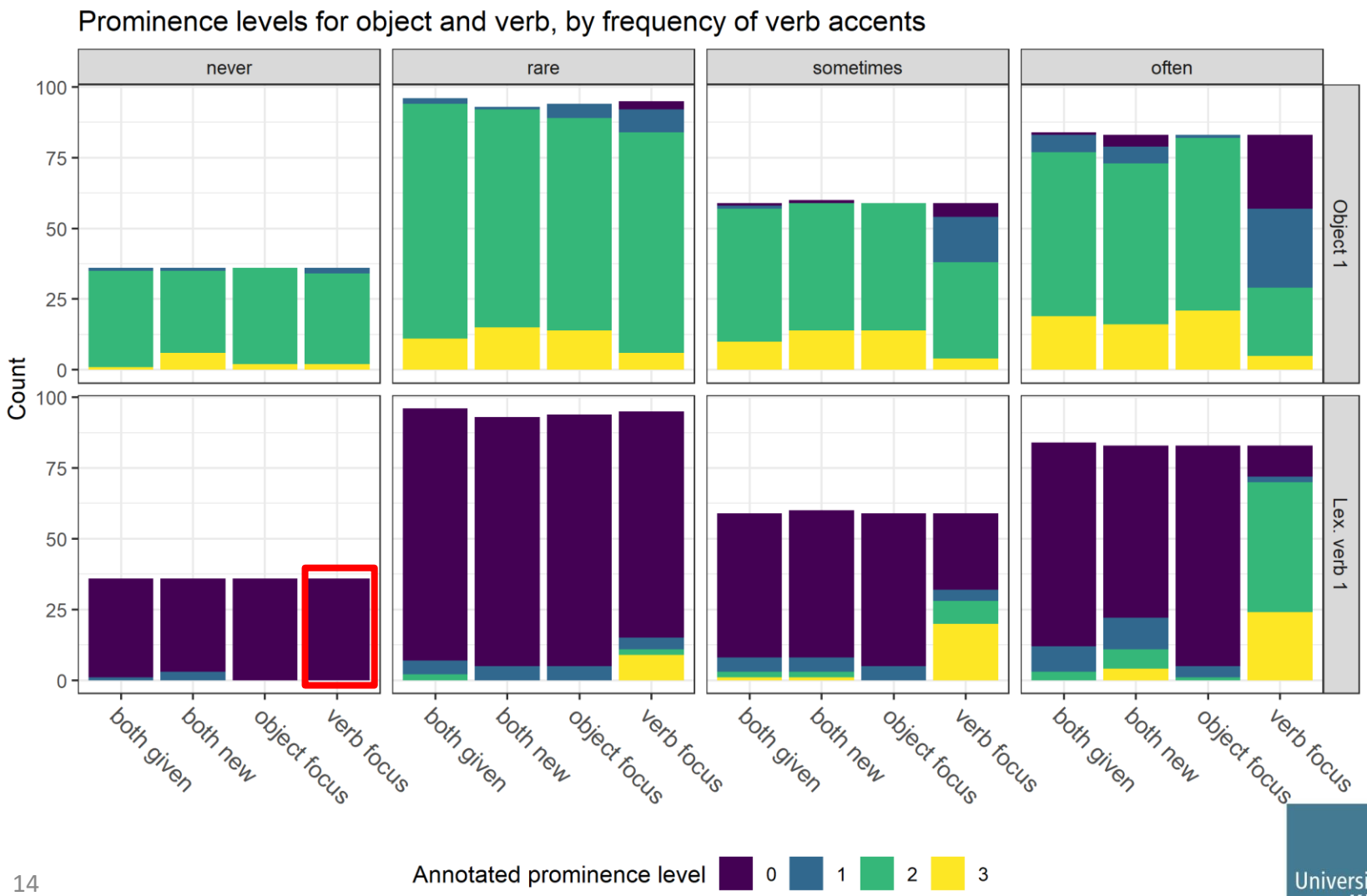

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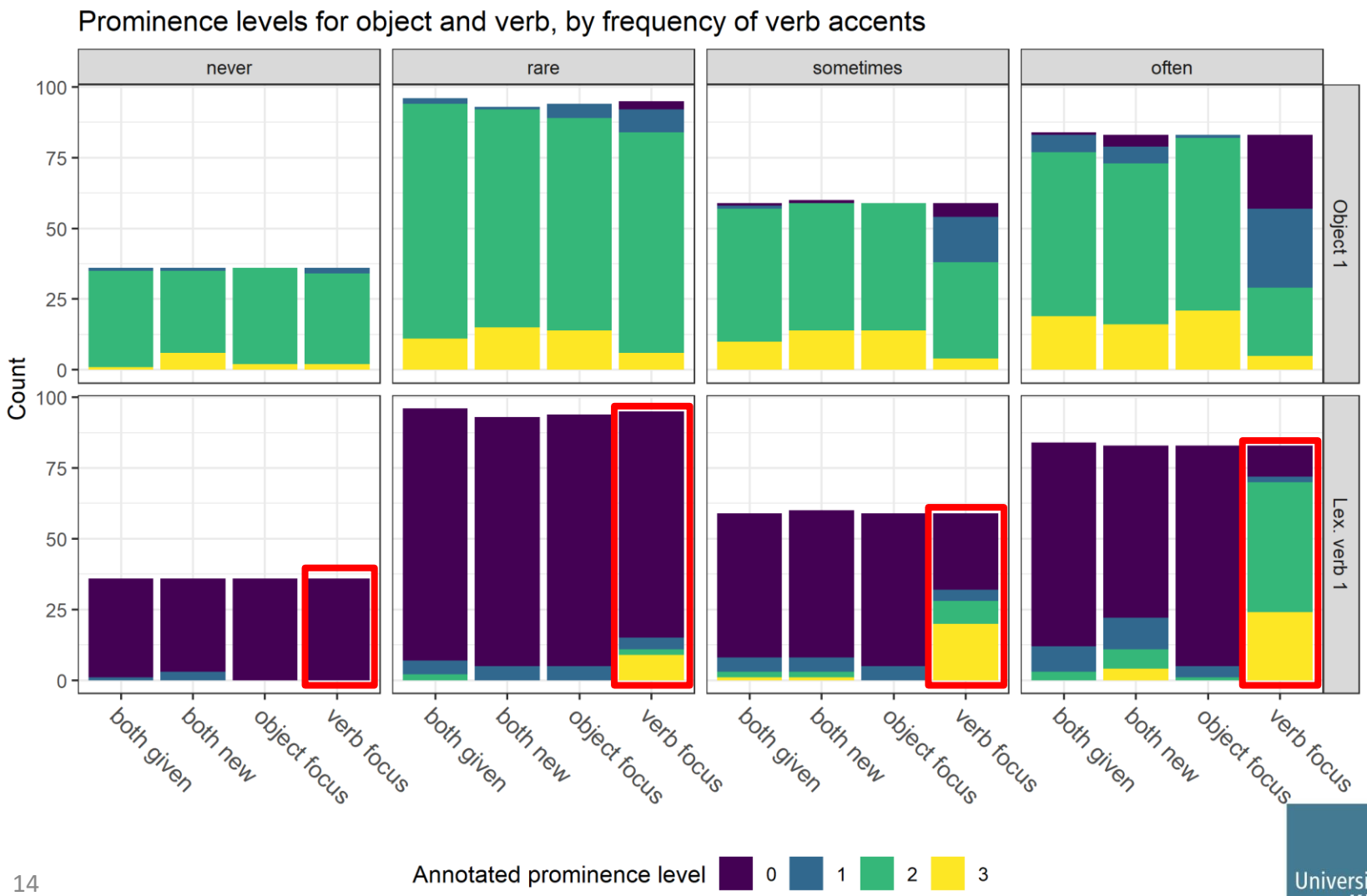

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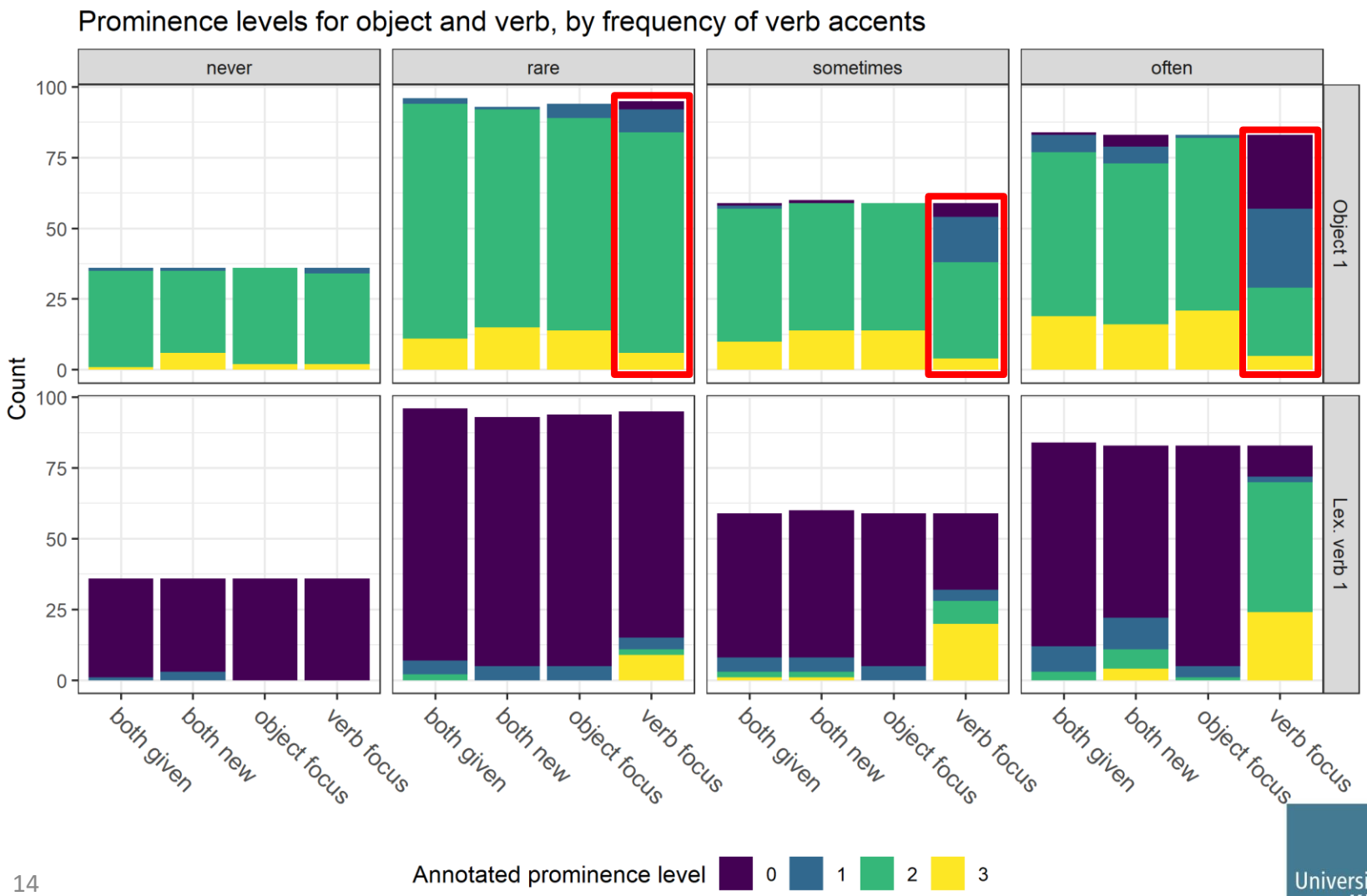

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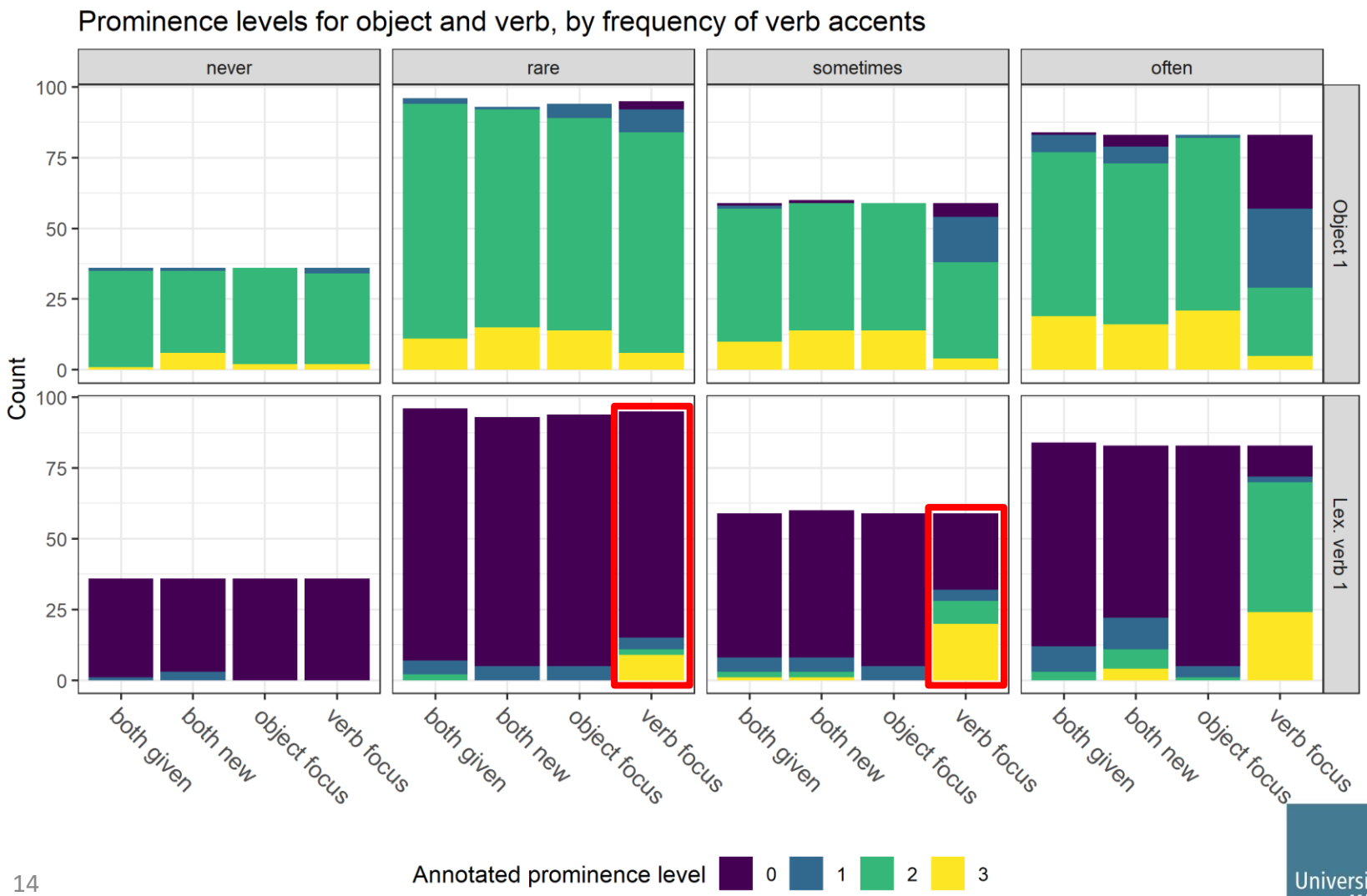

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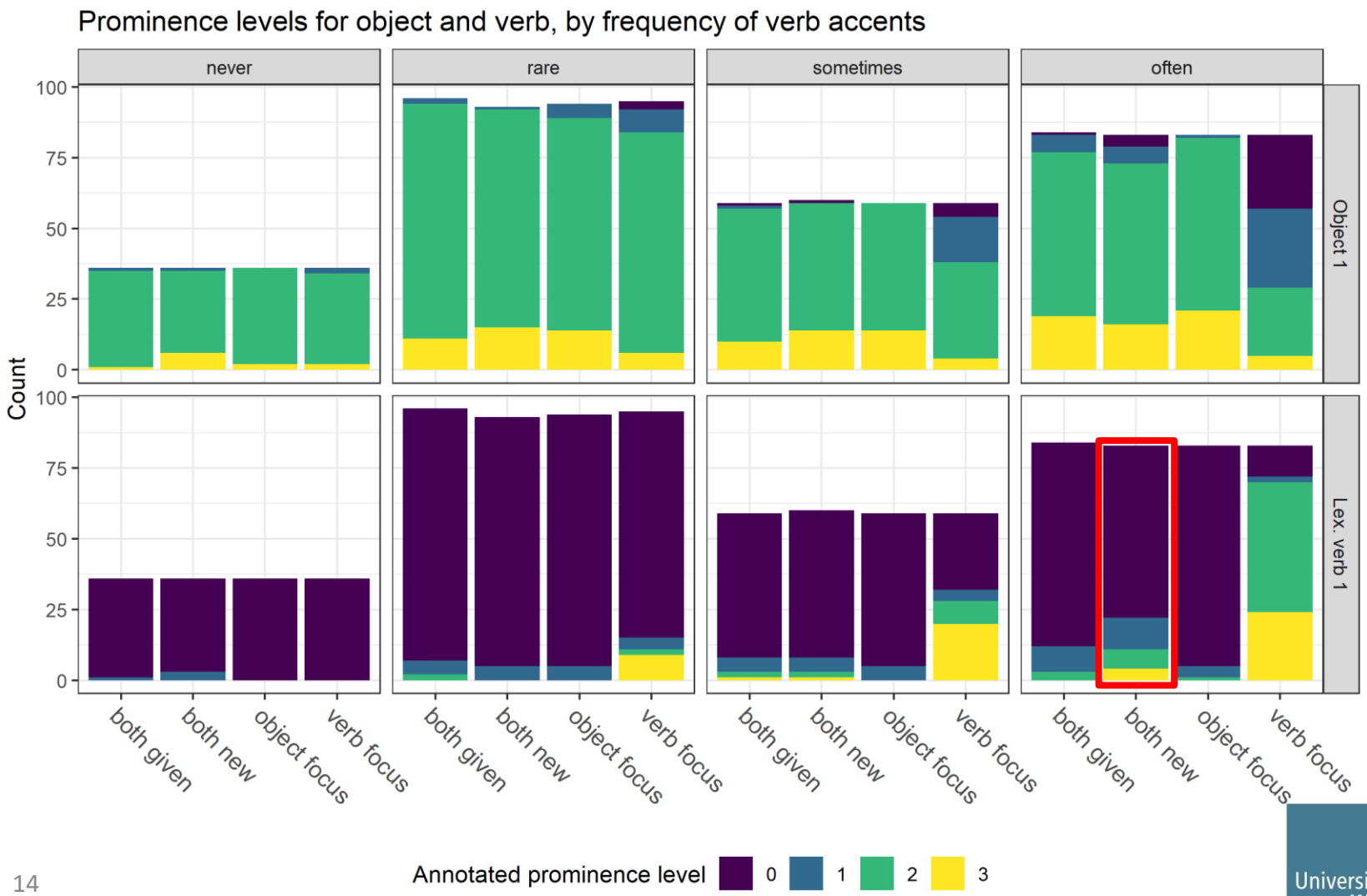

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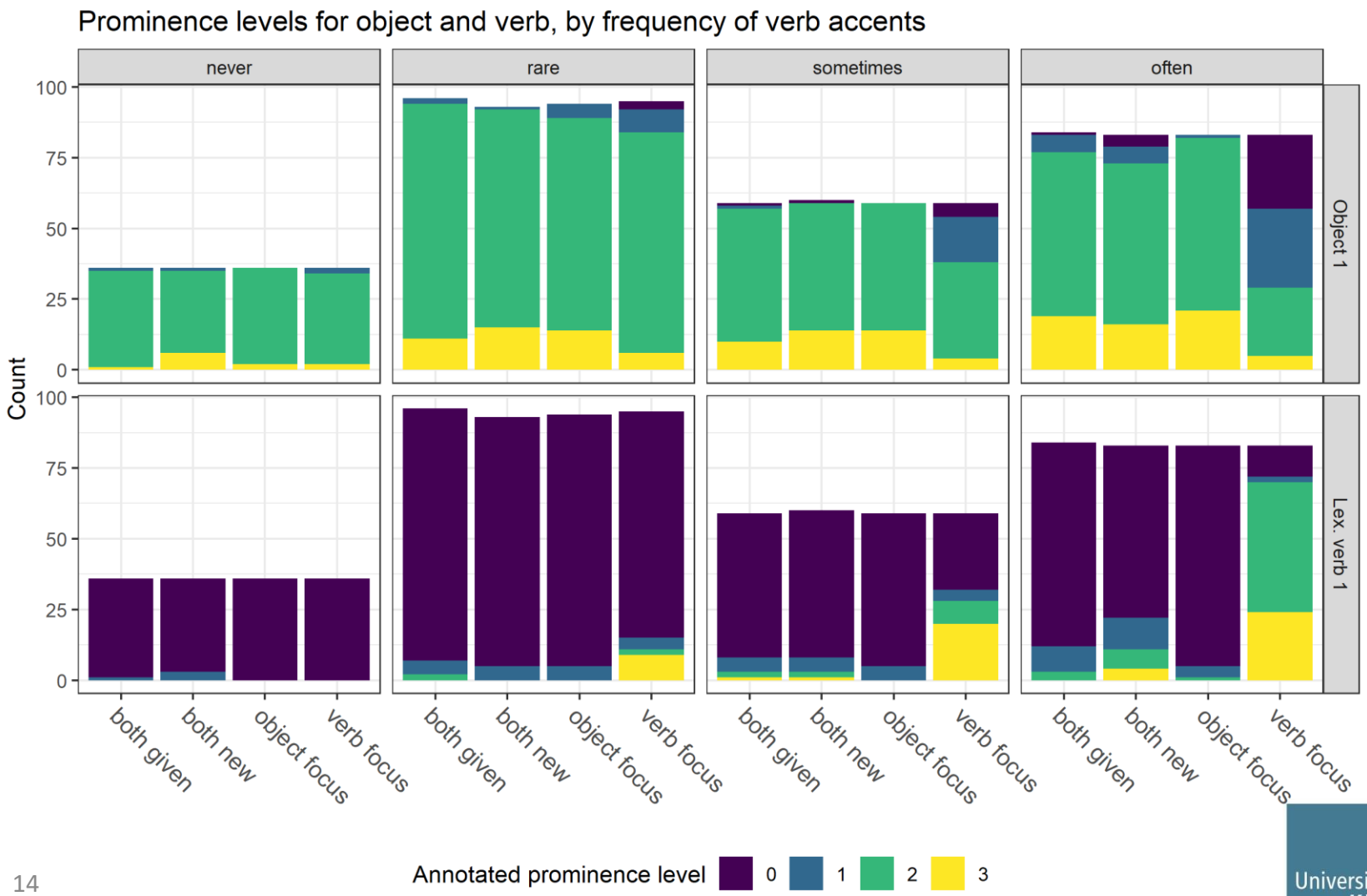

Broad vs. narrow focus

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Condition	Mean	SD
Both given	215.7	(56.6)
Both new	216.8	(54.4)
Object focus	224.9	(57.9)
Verb focus	210.8	(53)

Duration of stressed syllable of object in ms, when accented – all participants

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"This difference of 8 ms may be significant, but is it interesting?"

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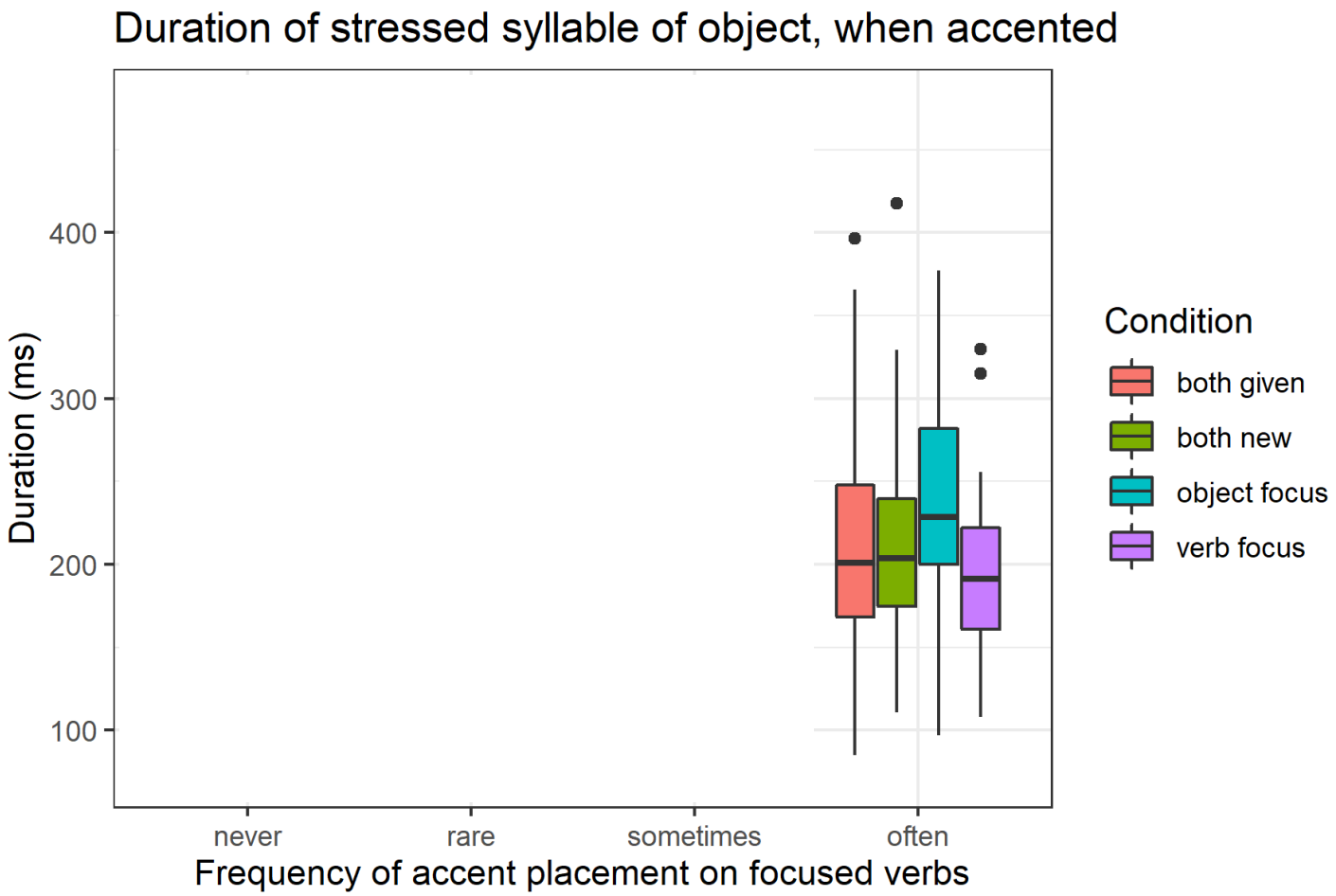

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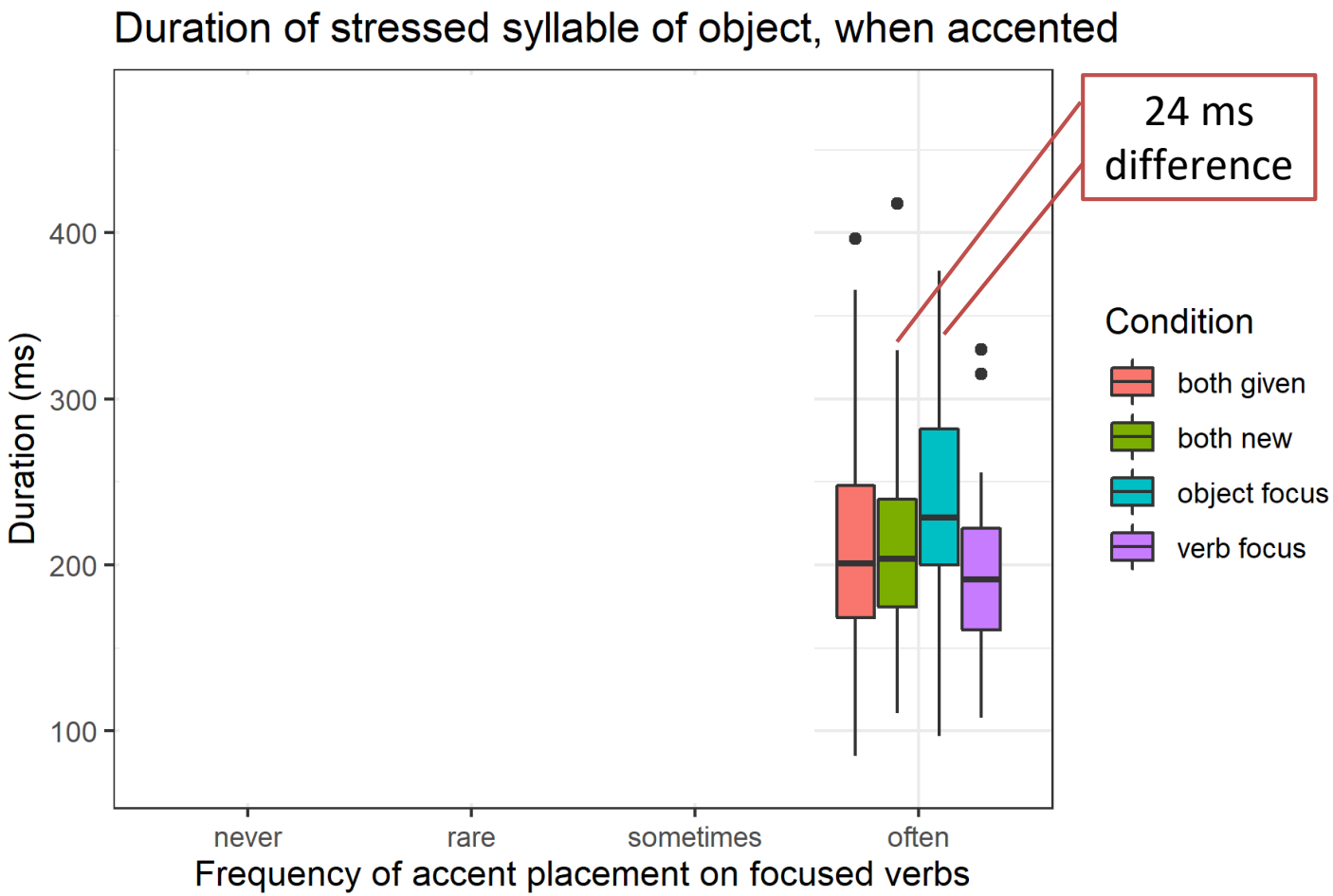

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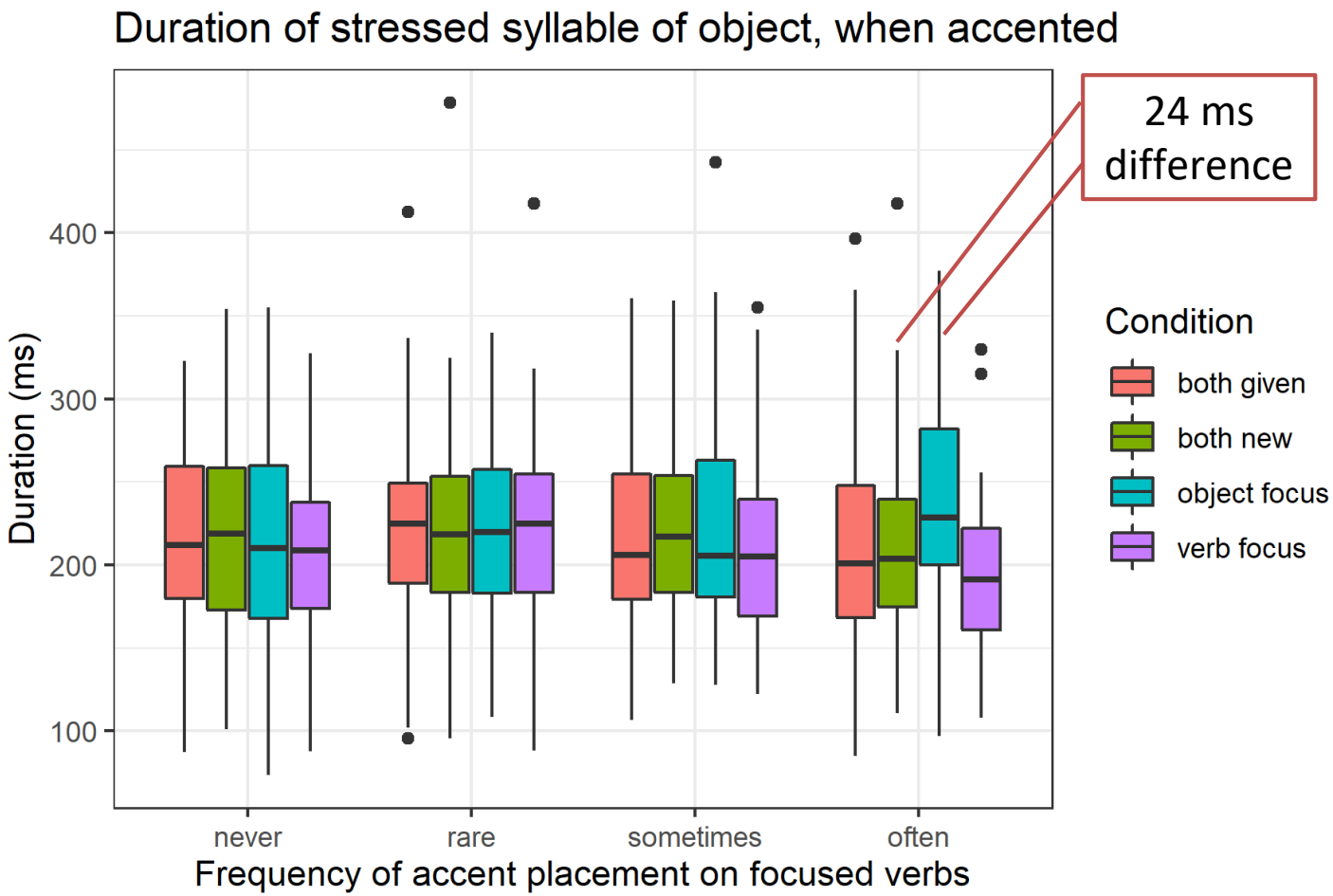

Possible explanations

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∼50% of participants being unaffected by a big experimental manipulation is a lot – so what happened?

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- Option A: Mapping of information structure to prosody is different in questions vs. assertions.
  - Not very likely (and what's the source of the variability?)

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Possible explanations

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~50% of participants being unaffected by a big experimental manipulation is a lot – so what happened?

- Option A: Mapping of information structure to prosody is different in questions vs. assertions.
  - Not very likely (and what's the source of the variability?)
- Option B: There were problems with the experimental design.
  - Possible (but what's the source of the variability?)

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Possible explanations

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~50% of participants being unaffected by a big experimental manipulation is a lot – so what happened?

- Option A: Mapping of information structure to prosody is different in questions vs. assertions.
  - Not very likely (and what's the source of the variability?)
- Option B: There were problems with the experimental design.
  - Possible (but what's the source of the variability?)
- Option C: Some participants read the target utterances with default prosody, detached from context.
  - Probably the likeliest.

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A note on the experimental design

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- Focus marking is actually probably harder to manipulate / elicit in questions than in assertions.
- E.g. the "both new" condition was probably a bit weird.
- Maybe some participants found this incoherent because a lot of accommodation is required.

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S2: Ja, das hat sie mir auch erzählt. Ich frage mich, was der Jäger ihr da alles zeigen wird. Jana war ja richtig heiß drauf, alles selbst auszuprobieren, stimmt's, Alex? Ich glaube, dass sie da dann auch SELBST Tiere erlegen will.
S3: Jana will da dann Rehe jagen?

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S3: Jana will da dann Rehe jagen?

Maybe accommodation difficulties make (some) participants "check out" mentally, even in other conditions?

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Possible solutions

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- Only keep data from participants who shifted the nuclear accent to the verb at least x% of the time; replace others until desired *n* is reached
  - Terrible idea:
  - Impossible to find unexpected results
  - Gives inaccurate description of sample if not reported completely
- Reporting on the whole data set, assigning participants to sub-groups with (hopefully) objective criteria
  - Less terrible idea – gives complete description of the data set, but criteria may not always be obvious

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Summary: linguistic results

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- There is decent evidence that speakers of German mark focus in declarative questions along these lines (if they do so at all):
  - Location of narrow focus shifts nuclear accent
  - Narrow focus: longer duration, larger pitch excursion (compared to broad focus)
  - Givenness tends to correlate with deaccentuation (but the object is very likely to carry a pre-nuclear accent even if given)

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  - Narrow focus: longer duration, larger pitch excursion (compared to broad focus)
  - Givenness tends to correlate with deaccentuation (but the object is very likely to carry a pre-nuclear accent even if given)
- No differences whatsoever for other speakers.

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Summary: methodological issues

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- My personal opinion: discarding data requires extremely good reasons (i.e. if in doubt, keep data).
- Subsetting data: OK in exploratory work, what about confirmatory research?
- If an experimental manipulation only works for 50% of participants: report that!
- It's valid and valuable information:
  - About a specific experimental design
  - Possibly about the research subject itself

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Thank you!

Any phonetic effects of focus marking?

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- Maybe the annotation is too coarse to account for phonetic/continuous focus marking?

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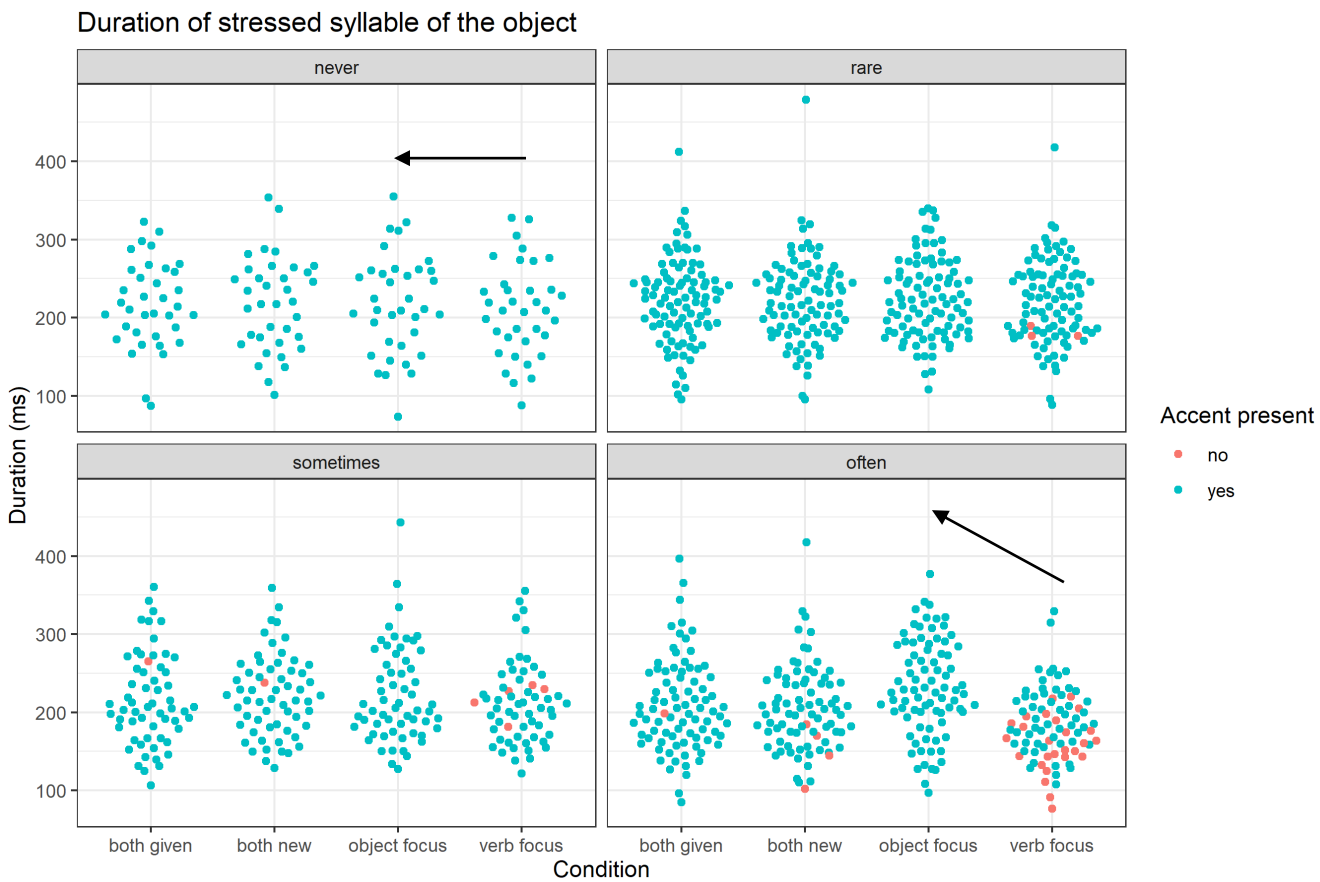

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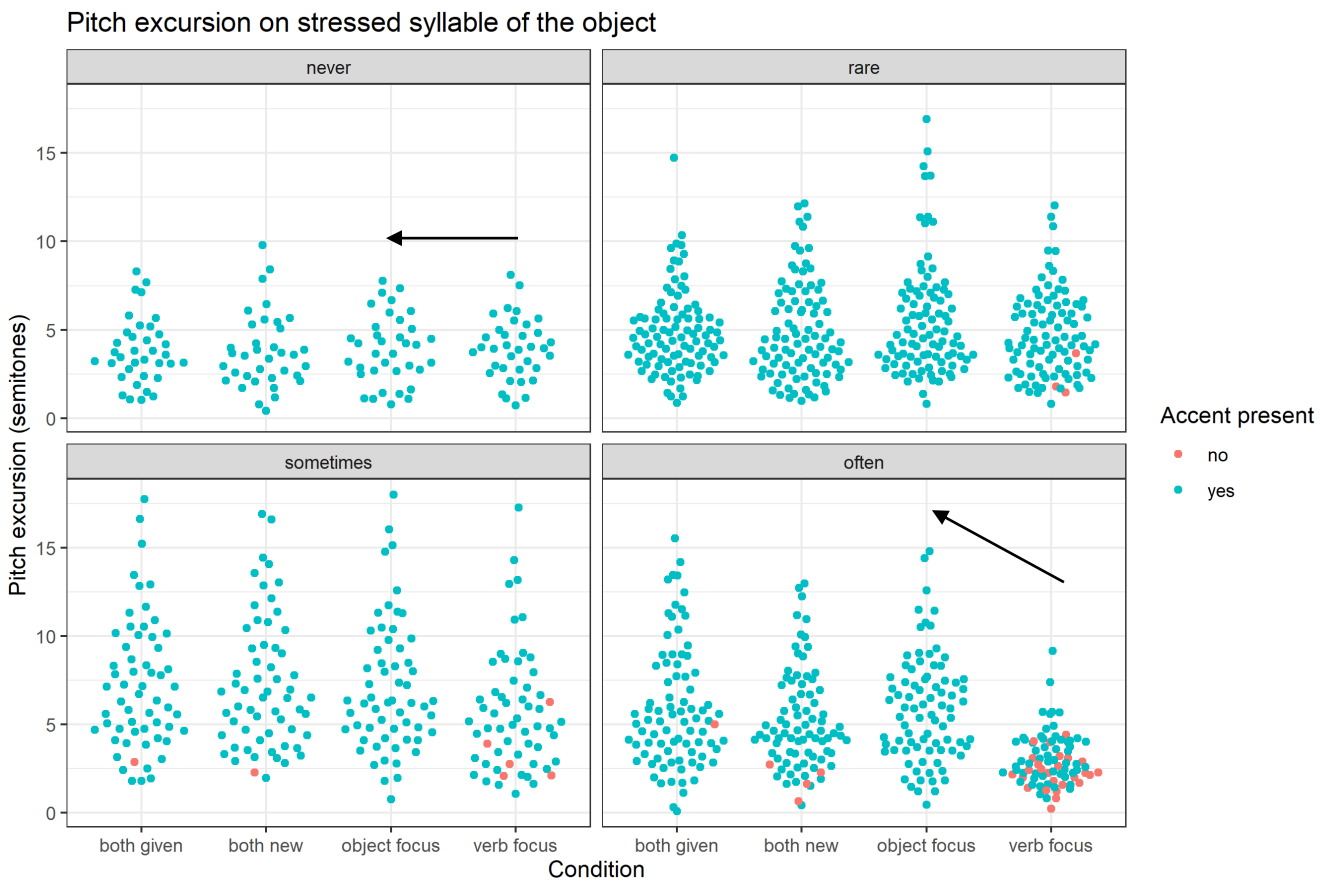

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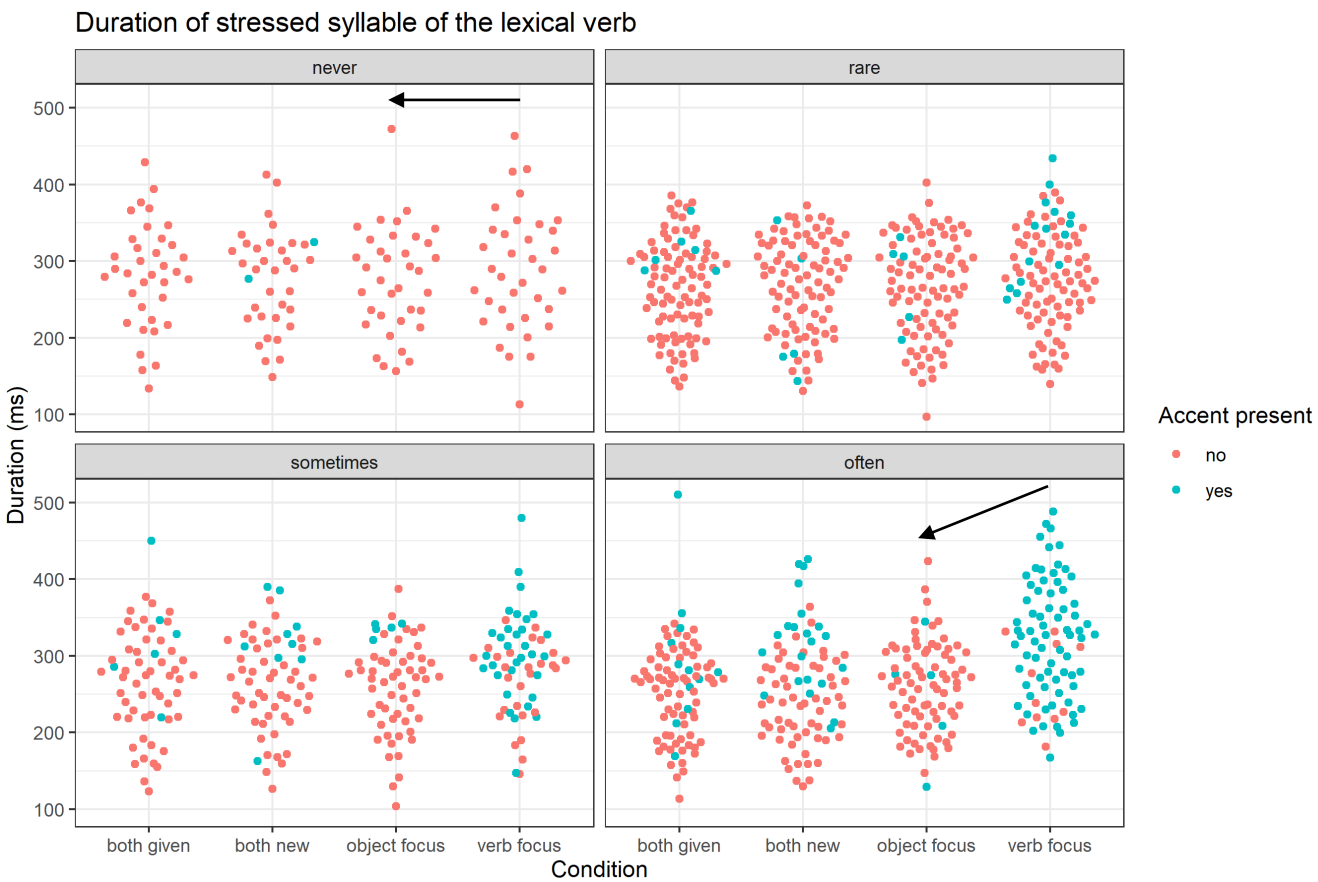

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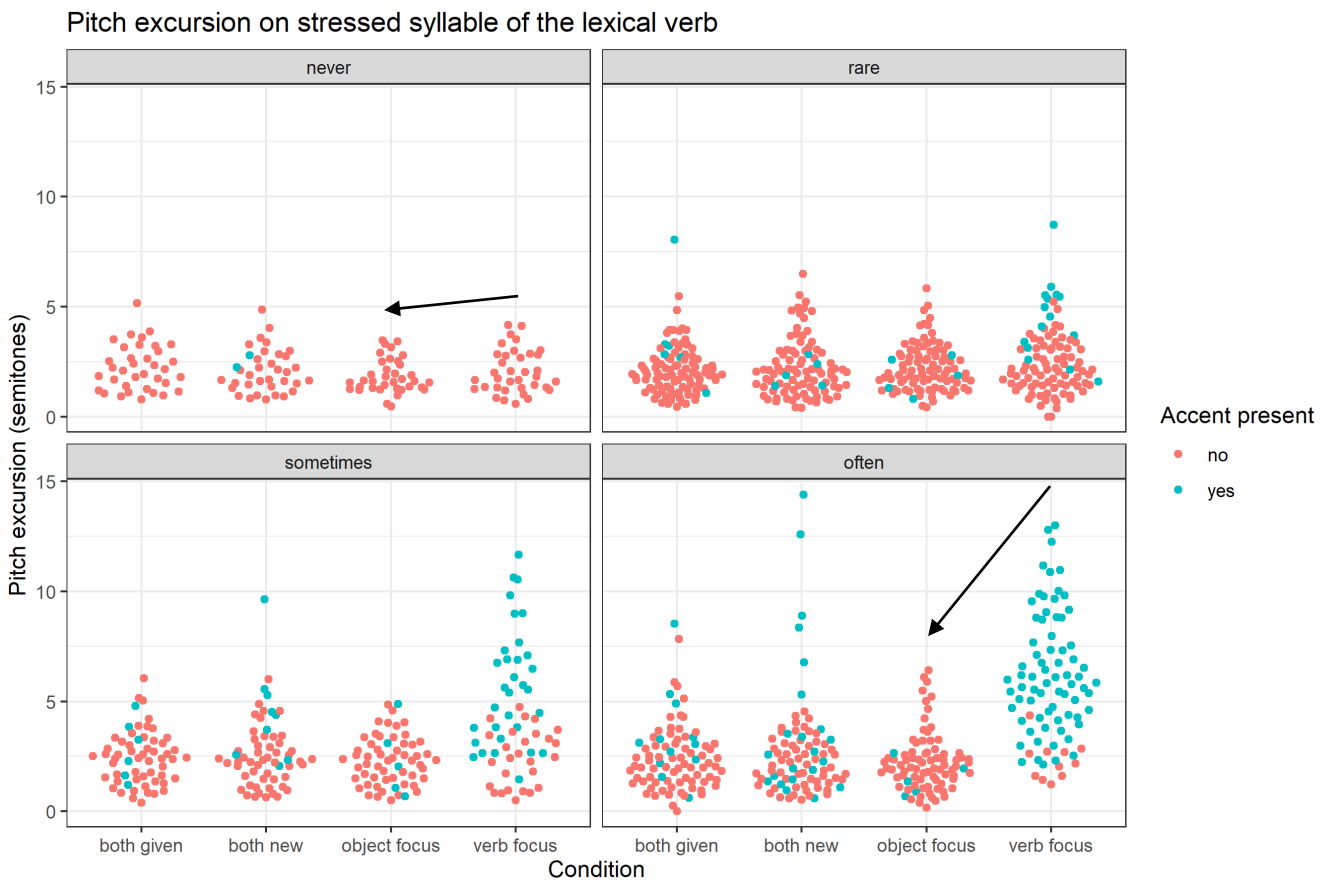

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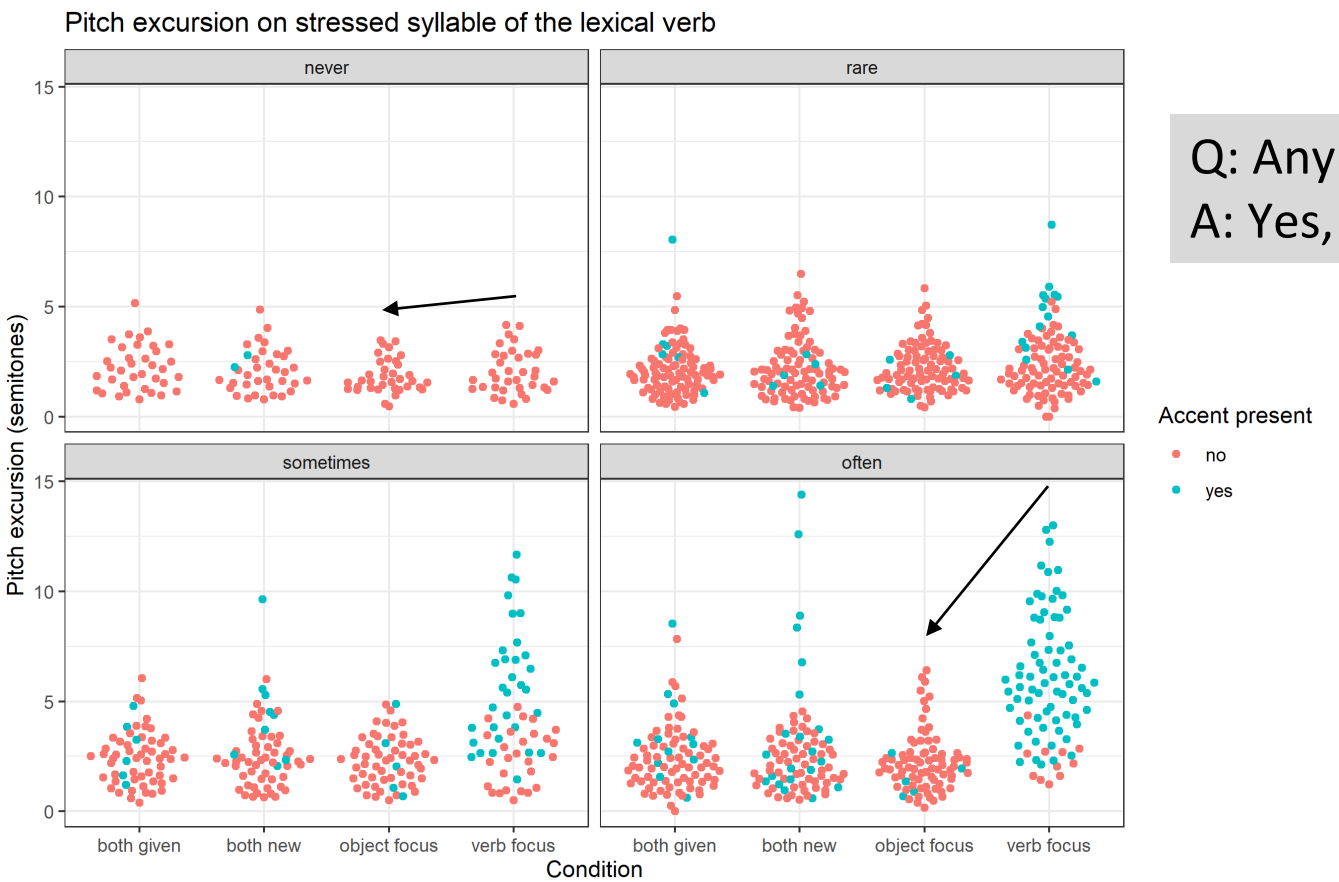

Q: Any phonetic effects?
A: Yes, but…

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Task description

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- When we ask participants to do X, what are we actually (not) telling them to do?
- "Ihre Aufgabe ist es, die Reaktion der dritten Person zu sprechen. Was genau Sie sagen sollen, ist vorgegeben. Bemühen Sie sich in jeden Fall, **möglichst natürlich** zu klingen. Seien Sie wissensdurstig, wenn Sie eine Frage stellen. Seien Sie aufgeregt, wenn Sie sich wundern, überrascht sind oder ähnliches. Stellen Sie sich also vor, Sie wären tatsächlich in dem Gespräch – und nicht in einem Labor."

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